

TrES-5b

R-0978

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230909 · created 2026-07-29T19:28:27+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460196.8243 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.052
Transit depth (Rp/Rs)^2	1.69 +/- 1.35 [%]
Orbital Inclination (inc)	80.84 +/- 2.8
Airmass coefficient 1 (a1)	1.6 +/- 0.86
Airmass coefficient 2 (a2)	-0.35 +/- 0.32
Scatter in the residuals of the lightcurve fit is	53.67 %
Best Comparison Star	#3 - [499, 161]
Optimal Aperture	3.0
Optimal Annulus	4.5
Transit Duration (day)	0.036 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.13 ± 0.052 vs 0.142 (deviation 0.21σ)
Duration fit vs published	0.036 d vs 0.0758 d (deviation 1.22σ)
Mid-time O–C	9.0 min
Depth SNR	2.3
Residual scatter	53.67 %

Light curve

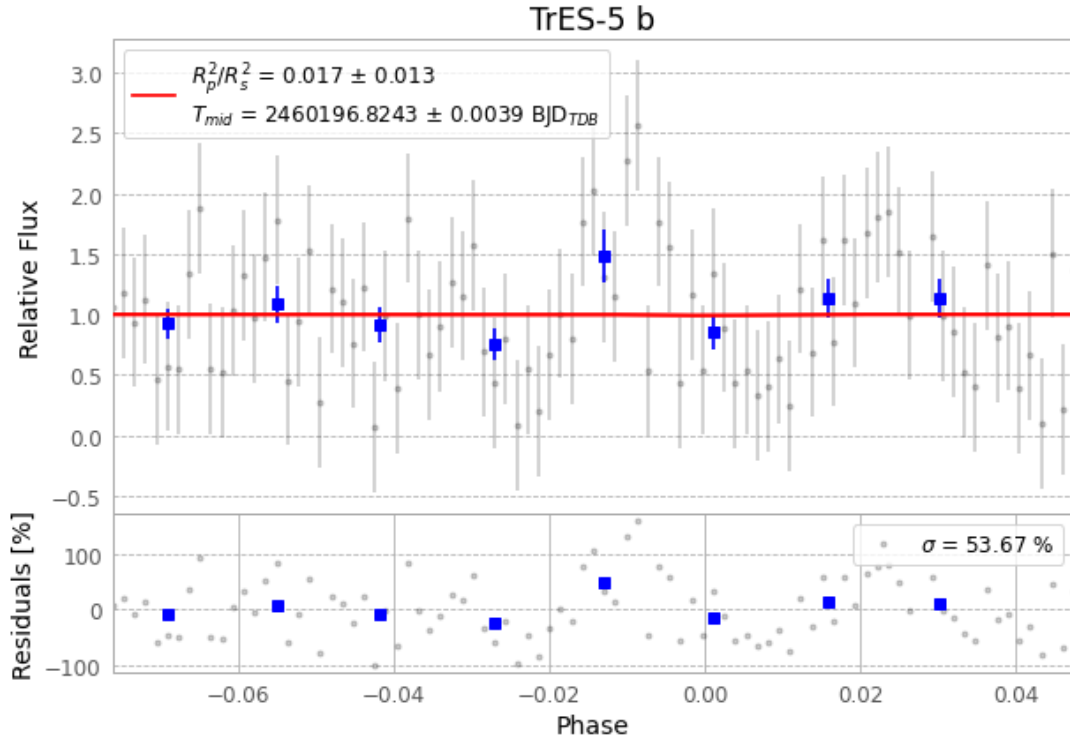


Figure 1 — FinalLightCurve_TrES-5 b_2023-09-08.png (SHA-256 bf231a92da960e15...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [532, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7bc2b490659edd60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

89 FITS frames (58 MB), TRES-5230909050015.FITS → TRES-5230909092416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230909.log (fa244b2f2bb1...), FinalLightCurve_TrES-5 b_2023-09-08.png (bf231a92da96...), FinalLightCurve_TrES-5 b_2023-09-08.csv (7a539005039a...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 19:28Z.